When Clinics Look Fine

but aren't…

Hidden Workload Under Partial Observability in Longitudinal Care

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PROBLEM

- In longitudinal care, **dashboard-visible** stability can **diverge from true burden**
- Routine metrics capture **scheduled and utilized** encounters more readily than
  - follow-up **obligations**
  - **hidden work**, and
  - **latent demand**
- A clinic can therefore look manageable while carrying substantially more burden than the dashboard can see

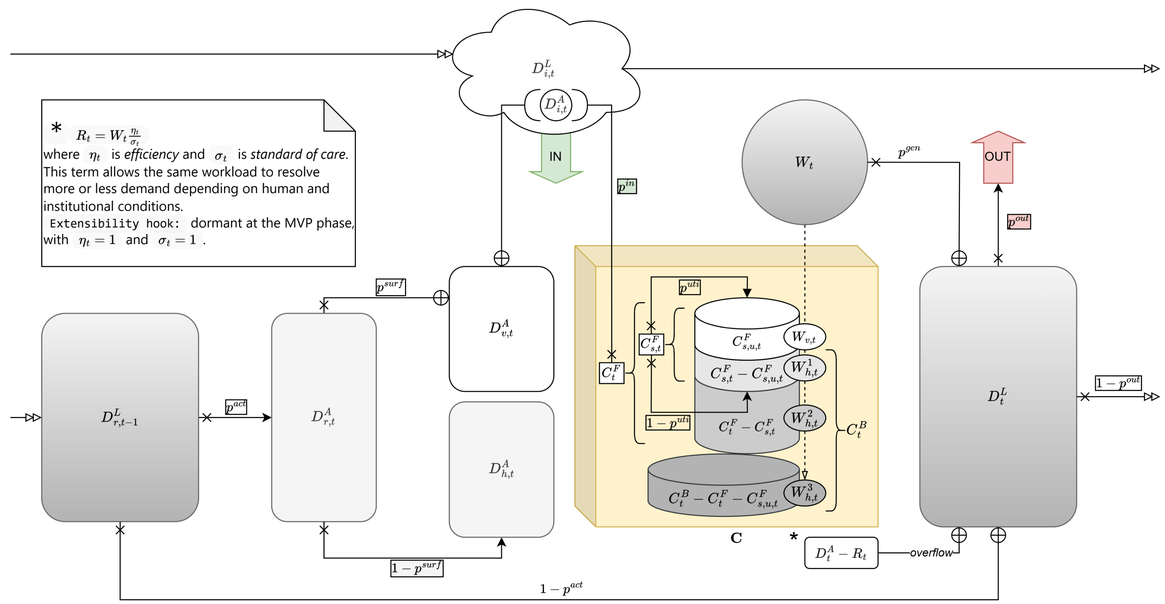
MODEL

- Simulation links **re-entrant demand**, **intake**, partial **observability**, formal **capacity**, and latent **carryover** across periods
  - p^{in} reserves intake capacity; p^{surf} determines what becomes visible; p^{act} activates pooled latent obligations; p^{out} determines whether burden clears or accumulates
- **Demand**, **capacity**, and **workload** remain distinct and are matched in *same unit (**minutes**)*
- The baseline is intentionally sparse: a minimal **sufficiency test** rather than a realism-heavy simulator

PRELIMINARY FINDINGS

- Increasing p^{surf} raises *visible* work and reduces apparent *slack*, with a modest decline in *overwork*
- Under **low observability**, slack-driven intake **expansion raises latent burden** above the fixed-intake baseline
- After reset, latent burden remains elevated, indicating **recovery lag**

Discrete-time stock-flow model



* $R_t = W_t \frac{\eta_t}{\sigma_t}$
where η_t is *efficiency* and σ_t is *standard of care*. This term allows the same workload to resolve more or less demand depending on human and institutional conditions.
`Extensibility hook:` dormant at the MVP phase, with $\eta_t = 1$ and $\sigma_t = 1$.

Fig 1. Ask me!

Notation Glossary			
D^L = Latent Demand	$\mathbf{C}$ = Maximal Capacity (bounded by period time-space physical limits)	W_v = Visible Workload	p^{act} = Activation Rate
D^A = Active Demand	C^F = Formal Capacity (template)	W_h = Hidden Workload	p^{surf} = Surfacing Rate
D_i = Intake Demand	C_s = Scheduled Formal Capacity	W^1_h = Through not utilized scheduled capacity	p^{uti} = Utilization Rate
D_r = Re-entrant Demand	$C_{s,u}$ = Utilized Formal Capacity	W^2_h = Through unscheduled capacity	$1-p^{uti}$ = No-show/cancel Rate
D_v = Visible Demand	C^B = Capacity Buffer	W^3_h = Throug human buffer	p^{in} = Access Rate
D_h = Hidden Demand			p^{out} = Attrition Rate

Dashboard View	
Future ATP	$\frac{C^F_{t+n} - C^F_{s,t+n}}{C^F_{t+n}}$
Apparent Slack	$\frac{C^F - C^F_{s,u}}{C^F}$
Utilization	$\frac{C^F_{s,u}}{C^F}$

Follow the evolving GitHub repository for model code, outputs, iterative refinement, and collaboration

Workload metrics by observability

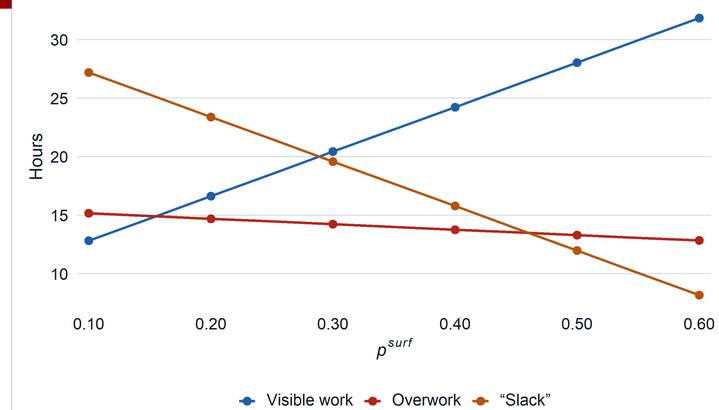


Fig 2. The same clinic can look **underfilled** or **overloaded** depending on what the dashboard can see. Those competing readings can drive **opposite operational policies**, especially around *access expansion*, *staffing*, and *protection of follow-up capacity*

Fixed vs slack-driven access policies

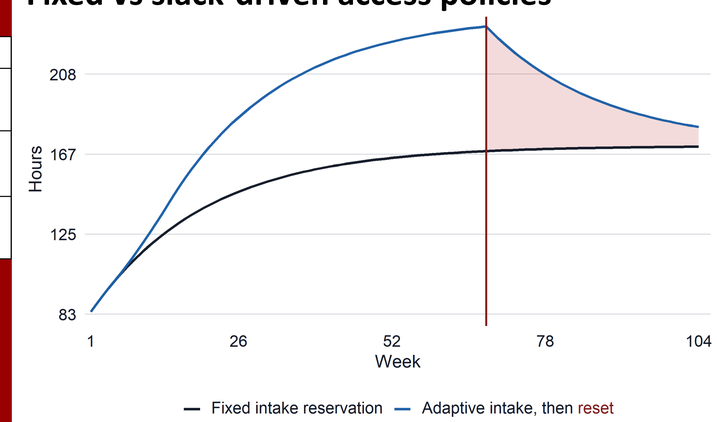


Fig 3. When apparent "**slack**" triggers **adaptive intake expansion**, *latent burden rises above the fixed-intake baseline* and stays elevated after reset. The remaining gap shows **recovery lag**: *policy reversal does not immediately undo the accumulated burden*